

Algebraic expressions

Letters standing for numbers — and the vocabulary the rest of the year needs.

LESSON 58

1 × 40 MIN

MATEMATIKA 7, §24

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

Words into symbols	10 min
The parts of an expression	12 min
Evaluating	12 min
Homework	6 min

LEARNING OBJECTIVES

- Write a worded statement as an algebraic expression.
- Name the terms, factors and coefficients of an expression.
- Evaluate an expression for given values of the letters.
- Distinguish an expression from an equation.

TEXTBOOK

National	pp. 152–155
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

Words into symbols

WORDS	EXPRESSION
five more than x	$x + 5$
five less than x	$x - 5$
five times x	$5x$
x divided by five	$\frac{x}{5}$
the square of x , less three	$x^2 - 3$
twice the sum of x and y	$2(x + y)$

“TWICE THE SUM” NEEDS BRACKETS

$2(x + y)$ and $2x + y$ are different expressions. The bracket is what makes the doubling apply to the whole sum.

The parts of an expression

An expression is built from **terms** separated by + and −. Each term is a product of a number and some letters.

EXPRESSION	TERMS	COEFFICIENTS
$3x + 5y - 7$	$3x, 5y, -7$	$3, 5$
$-4a^2 + a$	$-4a^2, a$	$-4, 1$
$x - \frac{y}{2}$	$x, -\frac{y}{2}$	$1, -\frac{1}{2}$

A term with no letter is a **constant**. A coefficient of 1 or −1 is not written: a means $1a$ and $-a$ means $-1a$.

THE SIGN BELONGS TO THE TERM

In $3x - 7y$ the second term is $-7y$, with coefficient -7 . Reading the coefficient as 7 loses the sign in every later calculation.

Evaluating

To **evaluate** an expression, substitute the given numbers and compute, respecting the order of operations.

EXPRESSION	$X = 3, Y = -2$	VALUE
$2x + y$	$6 + (-2)$	4
$x^2 - y$	$9 - (-2)$	11
xy	$3 \times (-2)$	-6
$2(x + y)$	$2(1)$	2

PUT NEGATIVE VALUES IN BRACKETS

Writing $x^2 - y$ as $9 - -2$ invites an error; $9 - (-2)$ does not. The brackets cost nothing and prevent the commonest slip in substitution.

02 Worked examples**EXAMPLE 1** Write as expressions: seven more than twice x ; the square of the sum of a and b .

- ① Twice x is $2x$.
- ② Seven more: $2x + 7$.
- ③ The sum is $a + b$.
- ④ Its square: $(a + b)^2$.
Brackets are essential.

ANSWER $2x + 7$ and $(a + b)^2$ **EXAMPLE 2** Name the terms and coefficients of $3x + 5y - 7$.

- ① Terms are separated by $+$ and $-$.
- ② $3x$, $5y$, -7 .
- ③ Coefficients: 3 and 5.
- ④ -7 is a constant term.

ANSWER

Three terms; coefficients 3 and 5

EXAMPLE 3 Evaluate $x^2 - y$ and $2(x + y)$ at $x = 3, y = -2$.

① $3^2 - (-2) = 9 + 2$
Brackets round the negative.

② $= 11$

③ $2(3 + (-2)) = 2(1)$

④ $= 2$

ANSWER**11 and 2****03** **Interactive model**

Read six phrases aloud and have the class write the expressions before any of them is discussed; the errors that appear are exactly the ones to teach.

This model could not start: den is not a function

04 **Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic expression	Algebraik ifoda	Алгебраическое выражение
Variable	O'zgaruvchi	Переменная
Term	Had	Член
Coefficient	Koeffitsiyent	Коэффициент
Constant	O'zgarmas	Постоянная
To evaluate	Qiymatini topmoq	Вычислить значение
To substitute	O'rniga qo'yimoq	Подставить
Equation	Tenglama	Уравнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Five less than x ” is:

“Twice the sum of x and y ” is:

How many terms in $3x + 5y - 7$?

The coefficient of y in $3x - 7y$:

7

-7

3

1

$x^2 - y$ at $x = 3, y = -2$:

7

11

-11

5

A term with no letter is:

a coefficient

a constant

a variable

an equation

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Five more than x

ANSWER $x + 5$

2. Five less than x

ANSWER $x - 5$

3. Five times x

ANSWER $5x$

4. x divided by five

ANSWER $\frac{x}{5}$

5. Terms in $3x + 5y - 7$

ANSWER 3

6. Coefficient of x in $3x$

ANSWER 3

7. Coefficient of a in $-a$

ANSWER -1

Medium · the standard question

7 PROBLEMS

1. Twice the sum of x and y

ANSWER $2(x + y)$

2. Seven more than twice x

ANSWER $2x + 7$

3. The square of the sum of a and b

ANSWER $(a + b)^2$

4. Coefficient of y in $3x - 7y$

ANSWER -7

5. $2x + y$ at $x = 3, y = -2$

ANSWER 4

6. $x^2 - y$ at the same

ANSWER 11

7. xy at the same

ANSWER -6

Hard · stretch and reason

7 PROBLEMS

1. $x^2 + 2xy + y^2$ at $x = 3, y = -2$

ANSWER 1

2. $\frac{x+y}{x-y}$ at $x = 3, y = -2$

ANSWER $\frac{1}{5}$

3. $-a^2$ at $a = -4$

ANSWER -16

4. $(-a)^2$ at $a = -4$

ANSWER 16

5. Write: the sum of the squares of a and b

ANSWER $a^2 + b^2$

6. Write: the square of the sum of a and b

ANSWER $(a + b)^2$

7. Are they the same?

ANSWER No — try $a = b = 1$

07 Homework

Homework — 5 tasks

Put every substituted negative value in brackets.

- Write as expressions: four less than three times x ; half the sum of a and b .
- Name the terms and coefficients of $5a - 2b + 9$.
- Evaluate $3x - 2y$ at $x = -1, y = 4$.
- Evaluate $(x + y)^2$ at the same values.

5. Explain the difference between $a^2 + b^2$ and $(a + b)^2$.